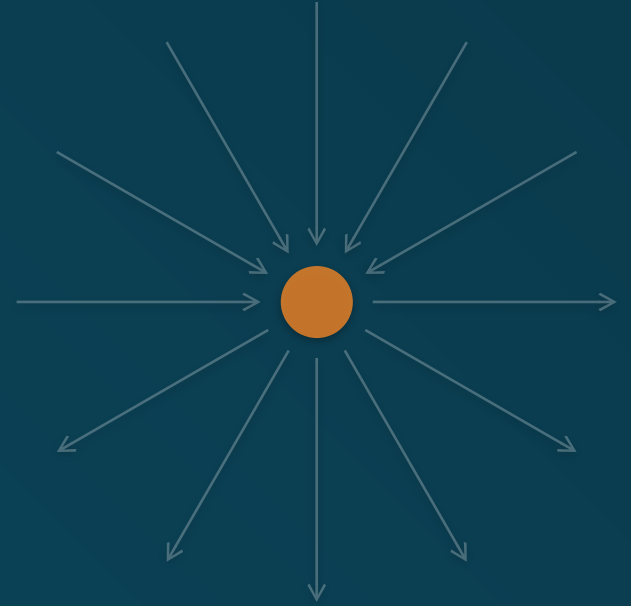


TOPIC 23

Nuclear Physics

核物理



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

23.1 Mass defect and nuclear binding energy 质量亏损与结合能

- Represent simple nuclear reactions with nuclide notation 核反应方程
- Mass-energy equivalence $E = mc^2$ 质能方程
- Mass defect, binding energy and binding energy per nucleon 质量亏损与结合能
- Sketch and interpret the BE-per-nucleon curve; fission and fusion 曲线解释

23.2 Radioactive decay 放射性衰变

- Spontaneous and random nature of decay 自发与随机
- Activity $A = \lambda N$ and the decay constant λ 活度与衰变常数
- Half-life $t_{1/2}$, $\lambda = 0.693/t_{1/2}$, $x = x_0 e^{(-\lambda t)}$ 半衰期与指数衰变

Mass–Energy Equivalence

质能等价

Einstein showed that mass and energy are equivalent: any change in energy is accompanied by a change in mass. In nuclear processes the mass changes are large enough to measure.

质量与能量等价：能量变化伴随质量变化。

$$E = mc^2$$

1 kg 9.0×10^{16} J

- Atomic mass unit: kg ; 能量等价 **1 u 934 MeV**
- 1 MeV = 1.60×10^{-13} J — the practical unit for nuclear energies 核物理常用单位
- Nuclear equations: conserve nucleon number (A) and proton number (Z) 配平 A 与 Z

Mass Defect & Binding Energy

质量亏损与结合能

The **mass defect** of a nucleus is the difference between the total mass of the separate nucleons and the mass of the nucleus.

质量亏损 = 独立核子总质量 - 原子核实际质量。

The **binding energy** of a nucleus is the energy equivalent of its mass defect — the energy needed to separate all its nucleons to infinity.

结合能 = 质量亏损的能量等价，即把核子全部分开所需能量。

$$BE = \Delta mc^2$$

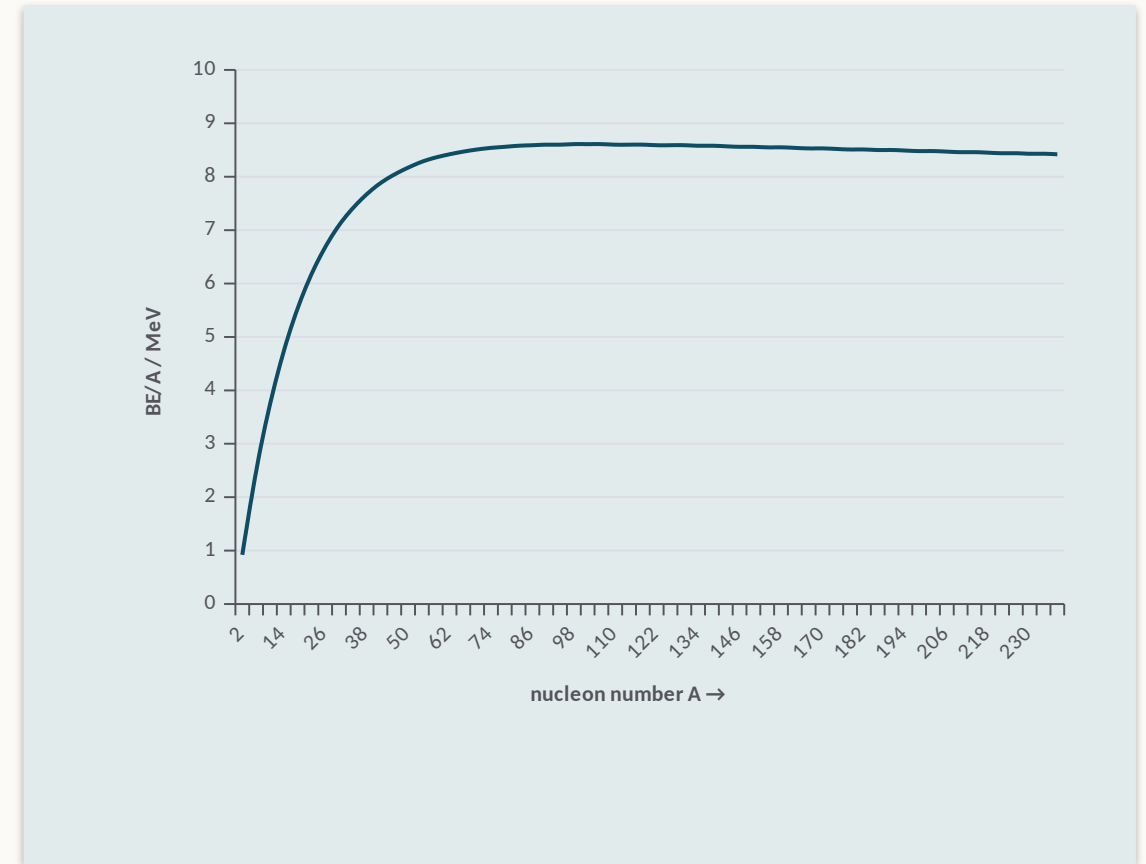
稳定核结合能大

- Binding energy is **released** when nucleons combine 核子聚合时释放
- Example: ${}^4\text{He}$ mass defect 0.0304 u $\Rightarrow$ BE $\approx$ 28.4 MeV 教材示例

Binding Energy per Nucleon

比结合能曲线

- BE per nucleon = $BE \div A$ — a measure of stability 比结合能越大越稳定
- Curve peaks near iron ($A \approx 56$) at ≈ 8.8 MeV 铁附近最大
- **Light** nuclei: fusion raises BE/A $\Rightarrow$ energy released 轻核聚变放能
- **Heavy** nuclei: fission raises BE/A $\Rightarrow$ energy released 重核裂变放能
- Both processes move products "up the curve" 产物都向峰移动



Variation of BE per nucleon with A 比结合能随质量数变化

Fission & Fusion

裂变与聚变对比

aspect 方面	nuclear fission 核裂变	nuclear fusion 核聚变
definition 定义	heavy nucleus splits into two lighter nuclei 重核分裂	two light nuclei join into a heavier nucleus 轻核聚合
typical reaction 典型反应	$^{235}\text{U} + \text{n} \rightarrow ^{141}\text{Ba} + ^{92}\text{Kr} + 3\text{n}$	$^2\text{H} + ^3\text{H} \rightarrow ^4\text{He} + \text{n}$ (in stars: $\text{H} \rightarrow \text{He}$)
condition 条件	triggered by neutron capture 中子轰击引发	very high T and pressure (thermonuclear) 高温高压
energy source 能量来源	nuclear power stations 核电站	the Sun and stars 太阳与恒星
by-products 产物	radioactive waste 放射性废料	far less radioactive waste 废料少
control 可控性	controlled chain reaction possible 可控链式反应	not yet sustained on Earth (JET / ITER) 尚未可控

- Chain reaction: each fission releases 2–3 neutrons which may trigger further fissions 链式反应
- Both raise the binding energy per nucleon of the products, releasing energy 都使产物比结合能增大

Random & Spontaneous Decay

衰变的随机性与自发性

Radioactive decay is **spontaneous**: it is unaffected by external conditions such as temperature or pressure. It is **random**: it is impossible to predict which nucleus will decay next, or when.

自发：不受外界条件影响；随机：无法预言哪个核、何时衰变。

- Evidence for randomness: count rate fluctuates about a mean value 计数率涨落是证据
- Each nucleus has a constant probability of decay per unit time 单位时间衰变概率恒定
- For a large number of nuclei, the decay follows a precise statistical (exponential) law 大量核服从统计规律
- The randomness of individual decays does not prevent accurate half-life measurements 个体随机，整体确定

Activity & Decay Constant

活度与衰变常数

The **activity** A of a source is the number of nuclear decays per unit time. The **decay constant** λ is the probability per unit time that a given nucleus will decay.

活度 = 单位时间衰变次数；衰变常数 = 单个核单位时间的衰变概率。

$$A = - \frac{dN}{dt} = \lambda N$$

单位：贝克勒 Bq = s⁻¹

- $A \propto N$: activity falls as the number of undecayed nuclei falls 活度随核数同步下降
- λ is constant for a given nuclide — large λ means rapid decay λ 大则衰变快
- Corrected count rate $\approx$ activity (background subtracted) 计数率需减本底

Half-Life

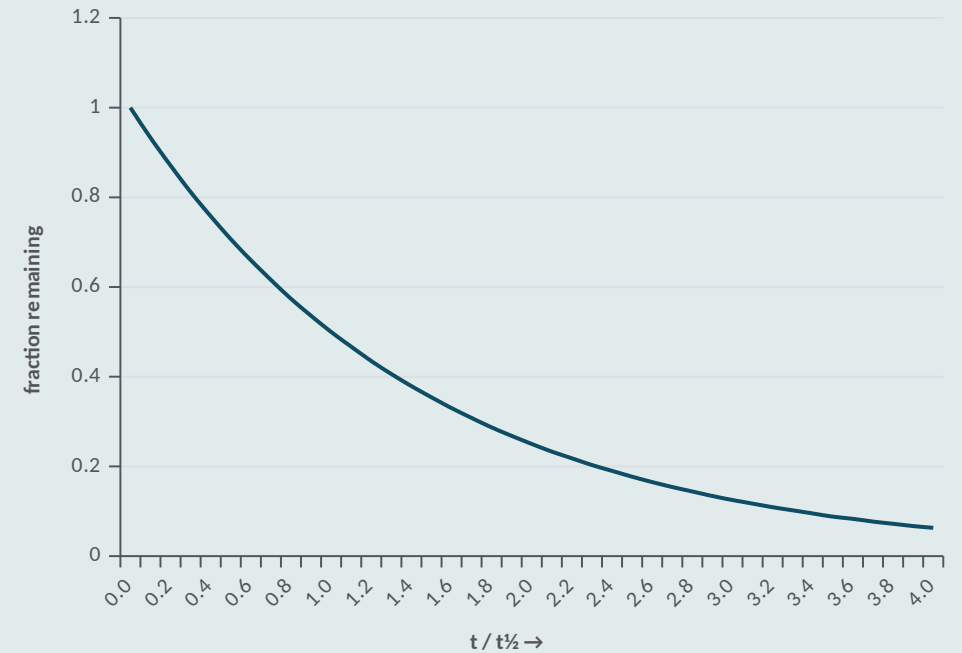
半衰期

The **half-life** $t_{1/2}$ of a radioactive nuclide is the time taken for half of the nuclei in a sample to decay.

半衰期：样品中一半原子核发生衰变所需的时间。

$$\lambda = \frac{0.693}{t_{1/2}}, x = x_0 e^{-\lambda t} \quad x \text{ 可以是 } N、A \text{ 或质量 } m$$

- After n half-lives the fraction remaining is $(1/2)^n$ n 个半衰期后剩 $(1/2)^n$
- Half-life is constant — independent of how many nuclei remain 与剩余核数无关
- Read $t_{1/2}$ from a decay curve at $N = N_0/2, N_0/4 \dots$ 从衰变曲线读取



Exponential decay curve 指数衰变曲线

Decay Calculations

三类典型计算

1. **λ from $t_{1/2}$:** $\lambda = 0.693/t_{1/2}$ — keep time units consistent (convert years to s) 单位统一
 2. **A or N from $A = \lambda N$:** e.g. mass of sample $\rightarrow$ number of atoms $N = m/M \times N_A$ 先求核数
 3. **Fraction remaining:** $x/x_0 = e^{(-\lambda t)} = (1/2)^{(t/t_{1/2})}$ 指数或用半衰期个数
- Carbon-14 dating: living matter keeps ^{14}C activity constant; after death it decays with $t_{1/2} = 5730 \text{ yr}$ 碳-14 定年
 - Compare measured activity with the "living" value to find the age 对比活度求年代
 - Check: does your final answer have sensible units and magnitude? 量纲与数量级自检

Worked Example 23A

例题解析

WORKED EXAMPLE 23A

Calculate the mass defect for a carbon-14 nucleus.

The measured mass is 14.003240 u.

(proton mass 1.007276 u; neutron mass 1.008665 u)

质量亏损 = 核子总质量 - 原子核质量，勿减反。

SOLUTION 解答

Step 1 ^{14}C has 6 protons and 8 neutrons

Step 2 Total mass of nucleons = $6 \times 1.007276 + 8 \times 1.008665 = 14.112976 \text{ u}$

Step 3 Mass defect = $14.112976 - 14.003240 = 0.109736 \text{ u}$

Answer $\Delta m = 0.1097 \text{ u}$

Worked Example 23B

例题解析

WORKED EXAMPLE 23B

Calculate the binding energy, in MeV, of a carbon-14 nucleus with a mass defect of 0.109736 u.

两种方法都要会：直接用 934 MeV/u 换算最快。

SOLUTION 解答

Step 1 Method 1: use 1 u 934 MeV: $BE = 0.109736 \times 934 \approx 102 \text{ MeV}$

Step 2 Method 2: $E = \Delta mc^2 = 0.109736 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 = 1.64 \times 10^{-11} \text{ J}$

Step 3 Convert: $1.64 \times 10^{-11} / 1.60 \times 10^{-13} \approx 102 \text{ MeV}$

Answer **$BE \approx 102 \text{ MeV}$**

Worked Example 23C

例题解析

WORKED EXAMPLE 23C

A nucleus of uranium-234 decays by α -emission to thorium. The total energy released is 4.77 MeV. Calculate the total change in mass during the decay.

质量减少，转化为 α 粒子与子核的动能。

SOLUTION 解答

Step 1 Convert energy: $4.77 \text{ MeV} = 4.77 \times 1.60 \times 10^{-13} = 7.63 \times 10^{-13} \text{ J}$

Step 2 $\Delta m = E/c^2 = 7.63 \times 10^{-13} / (3.00 \times 10^8)^2 = 8.5 \times 10^{-30} \text{ kg}$

Step 3 In u: $8.5 \times 10^{-30} / 1.66 \times 10^{-27} = 5.1 \times 10^{-3} \text{ u}$

Answer $\Delta m = 8.5 \times 10^{-30} \text{ kg} \approx 5.1 \times 10^{-3} \text{ u}$

Worked Examples 23D & 23E

例题解析

WORKED EXAMPLE 23D

- 1) The binding energy of a helium-4 nucleus is 28.4 MeV. Find BE per nucleon.
- 2) Masses: ^{238}U nucleus 237.9997 u, proton 1.00728 u, neutron 1.00867 u. Find BE per nucleon of ^{238}U .

- 1) $\text{BE}/A = 28.4/4 = 7.1 \text{ MeV}$
- 2) $\Delta m = (92 \times 1.00728 + 146 \times 1.00867) - 237.9997 = 1.93588 \text{ u}$
3. $\text{BE}/A = 1.93588 \times 934 / 238 \approx 7.60 \text{ MeV}$

Answer 7.1 MeV; 7.60 MeV per nucleon

WORKED EXAMPLE 23E

For the fission $^{235}\text{U} + \text{n} \rightarrow ^{141}\text{Ba} + ^{92}\text{Kr} + 3\text{n}$, masses/u: n 1.009, ^{235}U 235.123, ^{141}Ba 140.912, ^{92}Kr 91.913. Find a) the change in mass; b) the energy released per fission.

1. Reactants: $235.123 + 1.009 = 236.132 \text{ u}$; products: $140.912 + 91.913 + 3 \times 1.009 = 235.852 \text{ u}$
2. $\Delta m = 236.132 - 235.852 = 0.280 \text{ u}$
3. $E = 0.280 \times 934 \approx 261 \text{ MeV}$

Answer $\Delta m = 0.280 \text{ u}$; $E \approx 2.6 \times 10^2 \text{ MeV}$

Class Practice

随堂练习

- Q1** The binding energy of a nitrogen-14 nucleus is 101 MeV. Calculate its binding energy per nucleon.
- Q2** Cobalt-60 has a half-life of 5.3 years. Calculate the decay constant in s^{-1} , and the activity of a sample containing 2.0×10^{19} atoms.
- Q3** The activity of a radioactive source falls from 8.0×10^4 Bq to 5.0×10^3 Bq in 24 hours. Determine its half-life.

Answers Q1: $101/14 = 7.2$ MeV Q2: $\lambda = 4.1 \times 10^{-9} \text{ s}^{-1}$, $A = 8.3 \times 10^{10}$ Bq Q3: 比值 $16=2^4$, 4 个半衰期, $t_{1/2} = 6.0$ h

Summary

本章小结

KEY FORMULAS 核心公式

$$E = m c^2$$

质能方程

$$1 u \leftrightarrow 934 \text{ MeV}$$

质量 - 能量换算

$$B E = \Delta m c^2$$

结合能

$$A = \lambda N$$

活度

$$\lambda = 0.693 / t_{1/2}$$

衰变常数

$$x = x_0 e^{-\lambda t}$$

指数衰变

KEY IDEAS 核心思想

- Mass defect $\Rightarrow$ binding energy; larger BE/A $\Rightarrow$ more stable 质量亏损 $\Rightarrow$ 结合能
- Fusion of light and fission of heavy nuclei both release energy 聚变裂变皆放能
- Individual decays are random; large samples decay exponentially 个体随机，整体指数
- Keep u, kg, J, MeV conversions clear 单位换算链清晰

Exam Tips

备考提示

- Conversion chain: $u \rightarrow kg (\times 1.66 \times 10^{-27}) \rightarrow J (\times c^2) \rightarrow MeV (\div 1.60 \times 10^{-13})$. Or use $1 u = 934 MeV$ directly. 换算链别错。
- λ has unit s^{-1} : convert half-lives in years/days to seconds before using $A = \lambda N$. 时间单位统一到秒。
- For $e^{(-\lambda t)}$, compute the exponent first and check it is negative and dimensionless. 指数先算、必为负。
- Say "energy is released" for fission/fusion because products have higher BE/A — the direction of the statement earns marks. 表述方向：产物比结合能更大。